

HDOC DAY-19 ASSIGNMENT

Implementation of SGPA and CGPA Calculation Using Nested Loops & Decision-Making Constructs in C

1. Problem Understanding

The task is to implement the given algorithm for calculating Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA) for m semesters. Each semester contains n courses, and the number of courses may vary from one semester to another. The program must validate positive numeric inputs, accept only valid grades, convert grades to grade points, calculate semester totals, display each semester SGPA, and finally display the cumulative TC and CGPA.

- Use nested loops: an outer loop for semesters and an inner loop for courses.
- Validate `global_id`, m , n , course code, and credit so that only positive values are accepted.
- Accept grades O, A, B, C, D, E, or P and convert them to grade points 10, 9, 8, 7, 6, 5, and 4 respectively.
- Calculate SGPA for every semester and CGPA after all semesters are processed.

2. Standard Test Cases

Test Case	Complete Inputs	Expected Result	Status
TC1	<code>global_id = 101; m = 1</code> Semester 1: <code>n = 2</code> Course 1: <code>code = 101, credit = 4, grade = O</code> Course 2: <code>code = 102, credit = 4, grade = A</code>	Semester 1 SGPA = 9.50 CGPA = 9.50	Pass
TC2	<code>global_id = 205; m = 2</code> Semester 1: <code>n = 3</code> C1: <code>code = 101, credit = 4, grade = O</code> C2: <code>code = 102, credit = 3, grade = A</code> C3: <code>code = 103, credit = 3, grade = B</code> Semester 2: <code>n = 2</code> C1: <code>code = 201, credit = 4, grade = A</code> C2: <code>code = 202, credit = 4, grade = B</code>	Semester 1 SGPA = 9.10 Semester 2 SGPA = 8.50 CGPA = 8.83	Pass
TC3	<code>global_id = 0 (invalid), then 101; m = 1</code> Semester 1: <code>n = 1</code> Course 1: <code>code = 101, credit = 3, grade = B</code>	Error for <code>global_id</code> , then accept 101 Semester 1 SGPA = 8.00 CGPA = 8.00	Pass
TC4	<code>global_id = 101; m = 0 (invalid), then 1</code> Semester 1: <code>n = 1</code> Course 1: <code>code = 101, credit = 4, grade = C</code>	Error for m , then accept 1 Semester 1 SGPA = 7.00 CGPA = 7.00	Pass
TC5	<code>global_id = 101; m = 1</code> Semester 1: <code>n = 1</code> Course 1: <code>code = 101, credit = 3, grade = X (invalid), then O</code>	Error for grade X, then accept O Semester 1 SGPA = 10.00 CGPA = 10.00	Pass

3. Algorithm

1. Start.
2. Define the variables and their types: `global_id`, `m`, `n`, `semester`, `course`, `code`, `credit`, `GP`, `semester_TC`, `semester_TGC`, `cumulative_TC`, and `cumulative_TGC` as integers; `grade` as a character; and `SGPA` and `CGPA` as floating-point variables.
3. Get `global_id`.
4. Validate `global_id`. If `global_id` is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
5. Get `m`.
6. Validate `m`. If `m` is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
7. Set the cumulative values: `cumulative_TC = 0`, `cumulative_TGC = 0`.
8. Repeat Steps 8.1 to 8.8 for `m` semesters:
 - 8.1. Set the semester totals: `semester_TC = 0`, `semester_TGC = 0`.
 - 8.2. Get `n`.
 - 8.3. Validate `n`. If `n` is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
 - 8.4. Repeat Steps 8.4.1 to 8.4.9 for `n` courses:
 - 8.4.1. Get `code`.
 - 8.4.2. Validate `code`. If it is not positive, display an error message and request it again. Repeat until a positive value is entered.
 - 8.4.3. Get `credit`.
 - 8.4.4. Validate `credit`. If it is not positive, display an error message and request it again. Repeat until a positive value is entered.
 - 8.4.5. Get `grade`.
 - 8.4.6. Validate `grade`. Accept only O, A, B, C, D, E, or P. If the grade is invalid, display an error message and request it again.
 - 8.4.7. Find `GP` corresponding to grade: O = 10, A = 9, B = 8, C = 7, D = 6, E = 5, P = 4.
 - 8.4.8. `semester_TGC = semester_TGC + (GP * credit)`.
 - 8.4.9. `semester_TC = semester_TC + credit`.
 - 8.5. `SGPA = semester_TGC / semester_TC`.
 - 8.6. Display `semester` and `SGPA`.
 - 8.7. `cumulative_TGC = cumulative_TGC + semester_TGC`.
 - 8.8. `cumulative_TC = cumulative_TC + semester_TC`.
9. `CGPA = cumulative_TGC / cumulative_TC`.
10. Display `global_id`, `cumulative_TC`, and `CGPA`.
11. Stop.

4. C Program

```
#include <stdio.h>
int main()
{
    int global_id, m, n;
    int semester, course;
    int code, credit, GP;
    int semester_TC, semester_TGC;
    int cumulative_TC = 0, cumulative_TGC = 0;
    char grade;
    float SGPA, CGPA;

    printf("Enter global id: ");
    scanf("%d", &global_id);
    while(global_id <= 0) {
        printf("Invalid global id. Enter again: ");
        scanf("%d", &global_id);
    }

    printf("Enter number of semesters: ");
    scanf("%d", &m);
    while(m <= 0) {
        printf("Invalid value. Enter again: ");
        scanf("%d", &m);
    }

    for(semester = 1; semester <= m; semester++) {
        semester_TC = 0;
        semester_TGC = 0;

        printf("Enter number of courses in semester %d: ", semester);
        scanf("%d", &n);
        while(n <= 0) {
            printf("Invalid value. Enter again: ");
            scanf("%d", &n);
        }

        for(course = 1; course <= n; course++) {
            printf("Enter course code: ");
            scanf("%d", &code);
            while(code <= 0) {
                printf("Invalid code. Enter again: ");
                scanf("%d", &code);
            }

            printf("Enter course credit: ");
            scanf("%d", &credit);
            while(credit <= 0) {
                printf("Invalid credit. Enter again: ");
                scanf("%d", &credit);
            }

            printf("Enter grade: ");
            scanf(" %c", &grade);
            while(grade!='O' && grade!='A' && grade!='B' &&
                grade!='C' && grade!='D' && grade!='E' && grade!='P') {
                printf("Invalid grade. Enter again: ");
                scanf(" %c", &grade);
            }

            if(grade == 'O') GP = 10;
            else if(grade == 'A') GP = 9;
            else if(grade == 'B') GP = 8;
            else if(grade == 'C') GP = 7;
            else if(grade == 'D') GP = 6;
            else if(grade == 'E') GP = 5;
            else GP = 4;

            semester_TGC += GP * credit;
            semester_TC += credit;
        }

        SGPA = (float)semester_TGC / semester_TC;
    }
}
```

```
        printf("SGPA of semester %d = %.2f\n", semester, SGPA);
        cumulative_TGC += semester_TGC;
        cumulative_TC += semester_TC;
    }

    CGPA = (float)cumulative_TGC / cumulative_TC;
    printf("Global ID = %d\n", global_id);
    printf("Cumulative TC = %d\n", cumulative_TC);
    printf("CGPA = %.2f\n", CGPA);
    return 0;
}
```

5. Compilation and Execution Evidence

```
Enter grade: A
SGPA of semester 1 = 9.50
Global ID = 101
Cumulative TC = 8
CGPA = 9.50

(kali@kali)~/Desktop
$ ./question
Enter global id: 205
Enter number of semesters: 2
Enter number of courses in semester 1: 3
Enter course code: 101
Enter course credit: 4
Enter grade: 0
Enter course code: 102
Enter course credit: 3
Enter grade: A
Enter course code: 103
Enter course credit: 5
Enter grade: SGPA of semester 1 = 9.10
Enter number of courses in semester 2: 2
Enter course code: 201
Enter course credit: 4
Enter grade: A
Enter course code: 202
Enter course credit: 4
Enter grade: B
SGPA of semester 2 = 8.50
Global ID = 205
Cumulative TC = 18
```

```
Global ID = 205
Cumulative TC = 18
CGPA = 8.63

(kali@kali)~/Desktop
$ ./question
Enter global id: 0
Invalid global id. Enter again: 101
Enter number of semesters: 1
Enter number of courses in semester 1: 1
Enter course code: 101
Enter course credit: 3
Enter grade: B
SGPA of semester 1 = 8.00
Global ID = 101
Cumulative TC = 3
CGPA = 8.00

(kali@kali)~/Desktop
$ ./question
Enter global id: 101
Enter number of semesters: 0
Invalid value. Enter again: 1
Enter number of courses in semester 1: 1
Enter course code: 101
Enter course credit: 4
Enter grade: c
Invalid grade. Enter again: C
SGPA of semester 1 = 7.00
Global ID = 101
Cumulative TC = 4
CGPA = 7.00

(kali@kali)~/Desktop
$ ./question
Enter global id: 101
Enter number of semesters: 1
Enter number of courses in semester 1: 1
Enter course code: 101
Enter course credit: 3
Enter grade: X
```

```

kali@kali: ~/Desktop
Session Actions Edit View Help
Enter number of semesters: 1
Enter number of courses in semester 1: 1
Enter course code: 101
Enter course credit: 3
Enter grade: B
SGPA of semester 1 = 8.00
Global ID = 101
Cumulative TC = 3
CGPA = 8.00

kali@kali:~/Desktop$ ./question
Enter global id: 102
Enter number of semesters: 0
Invalid value. Enter again: 1
Enter number of courses in semester 1: 1
Enter course code: 101
Enter course credit: 4
Enter grade: C
Invalid grade. Enter again: C
SGPA of semester 1 = 7.00
Global ID = 101
Cumulative TC = 4
CGPA = 7.00

kali@kali:~/Desktop$ ./question
Enter global id: 102
Enter number of semesters: 1
Enter number of courses in semester 1: 1
Enter course code: 101
Enter course credit: 3
Enter grade: X
Invalid grade. Enter again: O
SGPA of semester 1 = 10.00
Global ID = 101
Cumulative TC = 3
CGPA = 10.00

kali@kali:~/Desktop$

```

6. Verification of Test Cases

Test Case	Complete Inputs	Expected Result	Actual Result	Status
TC1	global_id = 101; m = 1 Semester 1: n = 2 Course 1: code = 101, credit = 4, grade = O Course 2: code = 102, credit = 4, grade = A	Semester 1 SGPA = 9.50 CGPA = 9.50	Semester 1 SGPA = 9.50 CGPA = 9.50	Pass
TC2	global_id = 205; m = 2 Semester 1: n = 3 C1: code = 101, credit = 4, grade = O C2: code = 102, credit = 3, grade = A C3: code = 103, credit = 3, grade = B Semester 2: n = 2 C1: code = 201, credit = 4, grade = A C2: code = 202, credit = 4, grade = B	Semester 1 SGPA = 9.10 Semester 2 SGPA = 8.50 CGPA = 8.83	Semester 1 SGPA = 9.10 Semester 2 SGPA = 8.50 CGPA = 8.83	Pass
TC3	global_id = 0 (invalid), then 101; m = 1 Semester 1: n = 1 Course 1: code = 101, credit = 3, grade = B	Error for global_id, then accept 101 Semester 1 SGPA = 8.00 CGPA = 8.00	Error for global_id, then accept 101 Semester 1 SGPA = 8.00 CGPA = 8.00	Pass
TC4	global_id = 101; m = 0 (invalid), then 1 Semester 1: n = 1 Course 1: code = 101, credit = 4, grade = C	Error for m, then accept 1 Semester 1 SGPA = 7.00 CGPA = 7.00	Error for m, then accept 1 Semester 1 SGPA = 7.00 CGPA = 7.00	Pass
TC5	global_id = 101; m = 1 Semester 1: n = 1 Course 1: code = 101, credit = 3, grade = X (invalid), then O	Error for grade X, then accept O Semester 1 SGPA = 10.00 CGPA = 10.00	Error for grade X, then accept O Semester 1 SGPA = 10.00 CGPA = 10.00	Pass